

MD. ANISUR RAHMAN CHOWDHURY

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Authorized to work in the U.S. under F-1 OPT (STEM extension eligible) — no sponsorship required during OPT/STEM-OPT

PROFESSIONAL SUMMARY

Cloud, systems, and network solution architect with seven years spanning ISP operations, enterprise IT, and AWS security research, from a hundred-account deployment in Chittagong to graduate work at Gannon University. Built a serverless firewall that flags attack traffic at 98% accuracy, published across two IEEE conferences. CCNA, JNCIA, MTCNA certified; M.S., Information Technology, 3.95 GPA. Authorized under F-1 OPT, no sponsorship needed now, STEM-eligible later.

TECHNICAL SKILLS

Cloud & Architecture: AWS (EC2, VPC, CloudFront, S3, Lambda, WAF, IAM), serverless design, event-driven architecture, multi-cloud orchestration, DR planning, CI/CD, Proxmox, VMware ESXi, Hyper-V

Networking & Routing: BGP, OSPF, VLANs, STP, LACP, PoE, NAT, ACLs, TCP/IP, DNS, HTTP/S, TLS/SSL, REST APIs, SD-WAN, CDN, WAN topology design

Security: Zero Trust design, IDS/IPS, DDoS mitigation, RBAC, SIEM, incident response, OWASP, vulnerability management, firewall policy engineering

Systems & Services: Linux (RHEL, CentOS, AlmaLinux, Ubuntu), Windows Server, Active Directory, Group Policy, DHCP, Postfix, Dovecot, Apache, Nginx, LibreNMS

Automation & ITSM: Ansible, Terraform, PowerShell, ITIL-aligned incident and change management

Platform & Edge: Docker, Cloudflare (Tunnels, Workers, Pages, DNS), BIND9/PowerDNS, ZFS, Proxmox clustering

AI & Automation: LSTM, BiGRU, XGBoost, CNN, SVM, Ollama, n8n, FastAPI, MCP server development, prompt engineering, chain-of-thought pipelines

Development: Python, JavaScript, TypeScript, SQL, Java, Next.js, React, Tailwind CSS, Firebase, GitHub

PROFESSIONAL EXPERIENCE

IT Systems and Operations Engineer (Full-Time, OPT)

Jul 2026 – Present

Adel Group Inc, Winnetka, CA

- Replaced manual, per-device configuration with Ansible- and Terraform-based automation under ITIL-aligned change control, cutting configuration drift across IT systems and business applications.
- Monitor and maintain Layer 2 network infrastructure — VLANs, STP, LACP, PoE switching — catching and resolving incidents ahead of SLA breach.
- Tightened access control by enforcing RBAC policy, endpoint protection, and firewall rule hardening across the environment.
- Built AI-driven automation — LLM agents, n8n workflows, FastAPI services — to take routine IT tasks off the team's plate.
- Own vendor management and IT procurement strategy end to end.

Graduate Research Assistant — Cloud Security & AI Research

Aug 2025 – May 2026

Gannon University, CIS Department, Erie, PA

- Designed a serverless intelligent-firewall architecture on AWS Lambda, WAF, IAM, and S3 integrating LSTM, BiGRU, and XGBoost classifiers — 98% multi-class detection accuracy and 0.990 ROC-AUC across five attack categories on the CICIDS2017 benchmark, 135ms average response latency.
- Developed secure path-reservation algorithms and real-time traffic-flow models for enterprise-scale distributed networks; co-authored and presented the work at IEEE CSCloud 2025 and IEEE CSITSS 2025.
- Validated model reliability at 95% statistical confidence across three LLM benchmarks (MMLU, GSM8K, TruthfulQA) using paired t-tests and Wilcoxon signed-rank tests.

Cybersecurity Instructor — Applied Security Training (CPT)

Aug 2025 – Dec 2025

TAAS Services LLC, Remote

- Designed and taught 15+ hands-on lab environments on perimeter defense, threat analysis, and vulnerability management, mapped to MITRE ATT&CK and NIST frameworks.
- Led live penetration-testing simulations and incident-response drills; wrote the assessment rubrics and instructional materials from scratch.

Senior Executive, Information Technology

Dec 2021 – Jul 2024

S. Alam Group, Chittagong, Bangladesh

- Ran hybrid Linux (RHEL, CentOS, AlmaLinux) and Windows Server infrastructure for 500+ users, including Postfix/Dovecot mail, web hosting, and DNS/DHCP.
- Built out Active Directory identity infrastructure — GPO enforcement, RBAC, PowerShell-scripted provisioning — across multiple departments.
- Redesigned LAN/WAN topology, including a Layer 2 STP/LACP overhaul, and rolled out IP-PBX/VoIP under LibreNMS monitoring, cutting unplanned downtime.
- Led capacity planning, hardware procurement strategy, and technology vendor evaluation.

Technical Support Engineer, Networking

Aug 2017 – Dec 2021

Link3 Technology Ltd, Chittagong, Bangladesh

- Managed the full solution lifecycle — requirements scoping through POC deployment — for 100+ enterprise accounts across Cisco, MikroTik, Juniper, Fortinet, and Cyberoam platforms.
- Engineered site-to-site and remote-access IPsec/L2TP VPN architectures, firewall policy frameworks, NAT/ACL rule sets, and multi-link WAN failover with load balancing.
- Built standardized escalation runbooks for radio-link and fiber-optic transmission faults, cutting MTTR across dispersed enterprise sites under SLA pressure.

KEY PROJECTS AND RESEARCH

- **Serverless Intelligent Firewall (AWS):** Event-driven cloud security platform with automated threat response and Zero Trust policy enforcement; 135ms average response latency. Published IEEE CSCloud 2025.
- **Self-Correcting Reasoning Framework:** Three-stage Generator-Critic-Synthesizer pipeline for autonomous error detection and correction; 60% relative accuracy gain on GSM8K, 78% error correction rate.
- **Distributed Multi-Agent Ensemble System:** Orchestration layer coordinating four local inference agents using majority voting, dynamic weighted voting, and two-round debate strategies.
- **Active-Active DC/DR Lab:** High-availability Proxmox cluster with automated failover; production mail stack hardened with SPF, DKIM, and DMARC.

INDEPENDENT PROJECTS & INFRASTRUCTURE LAB

- **PentAI Pro:** LLM-driven automated penetration-testing platform across a three-node Proxmox deployment, with layered scope enforcement, tamper-evident audit logging, and prompt-injection defenses.
- **MCP Server Development:** Custom Model Context Protocol servers (Python FastMCP, Node.js SDK) exposing internal tools to LLM agents with scoped permissions and rate limiting.
- **DNS & Mail Infrastructure:** Self-hosted authoritative DNS on BIND9/PowerDNS with DNSSEC signing, alongside a Postfix/Dovecot mail stack with Sieve filtering and Rspamd spam mitigation.
- **Cloudflare Edge Platform:** Zero Trust tunneling to internal services without inbound ports, plus serverless functions on Workers and static deployment via Pages.
- **Containerized Services:** Docker-based service packaging and orchestration for lab workloads running alongside the Proxmox/ZFS storage tier.

EDUCATION

M.S., Information Technology — Gannon University, Erie, PA — GPA 3.95/4.00 — May 2026

B.S., Electronics and Telecommunication Engineering — University of Science and Technology Chittagong, Bangladesh — Jul 2017

CERTIFICATIONS

CCNA – Cisco Certified Network Associate (2024) | JNCIA – Juniper Networks Certified Associate (2022) | MTCNA – MikroTik Certified Network Associate (2019)

RESEARCH PUBLICATIONS (FIRST AUTHOR, IEEE)

- Towards a Serverless Intelligent Firewall: AI-Driven Security and Zero-Trust Architectures
- Towards a Serverless Intelligent Firewall: Integrating Cross-Cloud Adaptation, AI-Driven Security, and Zero-Trust Architectures
- FedThreat-X: A Privacy-Preserving Federated Threat Intelligence Framework for Multi-Cloud Cybersecurity
- Auction-Based Dynamic Resource Allocation for Optimized Edge Computing in Distributed Networks
- Teaching Large Language Models to Think Twice: A Three-Stage Self-Correction Framework
- A Local Distributed Multi-Agent LLM Ensemble System

Full publication list: [Google Scholar](#) | [ResearchGate](#)